

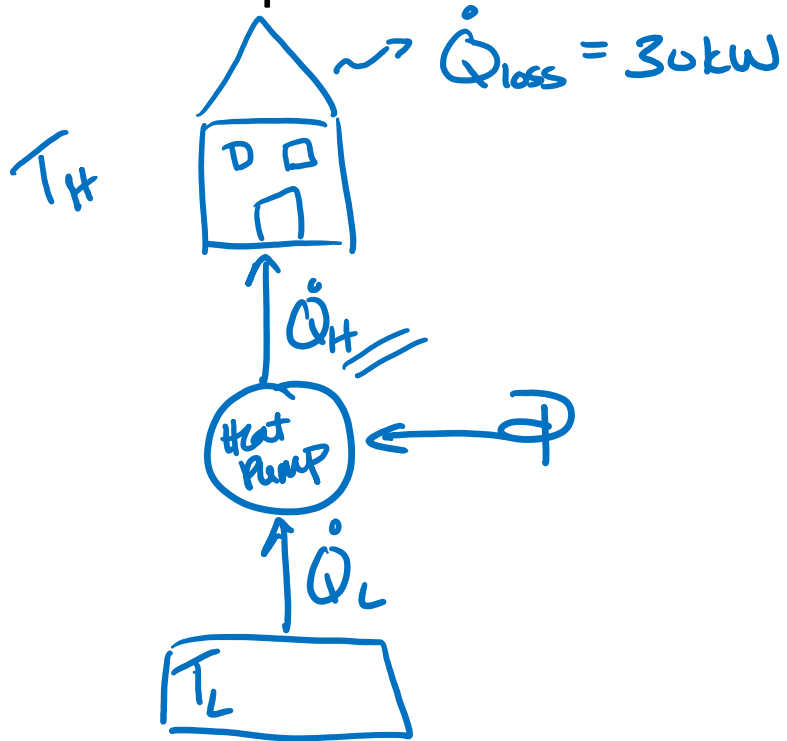


ME 300:
Engineering Thermodynamics

Heat Pumps and Refrigerators

Heat pump example

A heat pump is used to heat a house. The outside air is $T = -10^\circ\text{C}$ and the inside is a constant 20.3°C . The heat loss through the roof of the house is a steady 30 kW . Determine the minimum power required to run a heat pump to keep the house at this temperature.



$$T_H = 20.3^\circ\text{C} = 293.45\text{ K}$$

$$T_L = -10^\circ\text{C} = 263.15\text{ K}$$

$$T_H = \text{const} \Rightarrow \underline{\underline{\dot{Q}_H = \dot{Q}_{loss} = 30\text{ kW}}}$$



If we want the minimum power, what COP do we need?

- A. Minimum
- B. Maximum

Heat pump example

$$\underline{\underline{\beta = \frac{\dot{Q}_H}{\dot{Q}}}}} \quad \beta_{\max} = \beta_{\text{carnot}} = \frac{T_H}{T_H - T_L} = \frac{293.45}{293.45 - 263.15} = 9.68$$

$$\dot{Q} = \frac{\dot{Q}_H}{\beta_{\max}} = \frac{30 \text{ kW}}{9.68} \Rightarrow \boxed{\dot{Q}_{\min} = 3.1 \text{ kW}}$$